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PROJECT2: BUTTERWORTH LOW-PASS FILTER

1. Introduction

This report presents the analytical design of an 11th-order Butterworth low-pass filter (LPF) using the Sallen-Key topology. The design targets are a passband edge frequency of 14 kHz, a stopband edge of 20 kHz, a maximum passband ripple of 1 dB, and a minimum stopband attenuation of 25 dB.

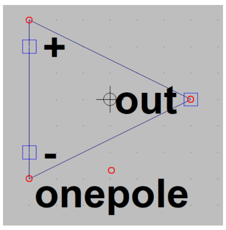
```
# Given specs
A_max_db = 1          # Amax = 1 dB
A_min_db = 25         # Amin = 25 dB
f_p = 14e3            # fp = 14 kHz
f_s = 20e3            # fs = 20 kHz
# GAIN = 8 V/V
# R TOL = 1%
# C TOL = 5%
```

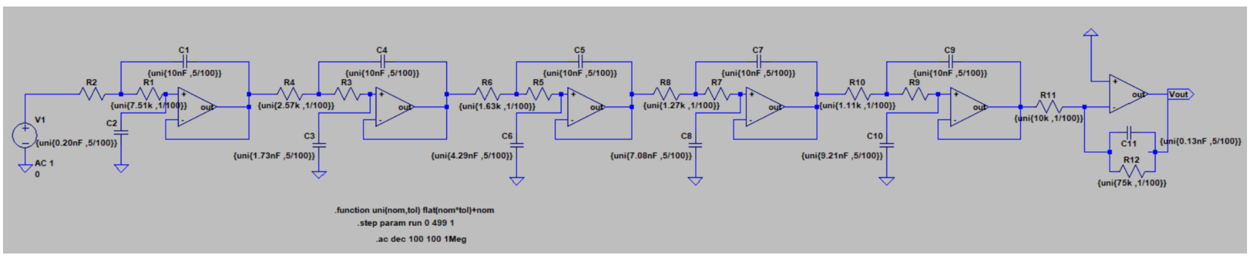
```
ws = 125663.70614359173
wp = 87964.5943005142
epsilon = 0.5088471399095875
for N = 11: A(ws) = 28.216730932810922
Pole circle radius = 93536.64 rad/s
```

```
Butterworth poles (11th order):
p1: -13311.65 + 92584.58j
p2: -38856.53 + 85083.92j
p3: -61253.48 + 70690.28j
p4: -78688.03 + 50569.73j
p5: -89747.75 + 26352.32j
p6: -93536.64 + 0.00j
p7: -89747.75 - 26352.32j
p8: -78688.03 - 50569.73j
p9: -61253.48 - 70690.28j
p10: -38856.53 - 85083.92j
p11: -13311.65 - 92584.58j
Q1 = 3.513    (w0 = 93536.64 rad/s)
Q2 = 1.204    (w0 = 93536.64 rad/s)
Q3 = 0.764    (w0 = 93536.64 rad/s)
Q4 = 0.594    (w0 = 93536.64 rad/s)
Q5 = 0.521    (w0 = 93536.64 rad/s)
```

```
Stage 1:
  Q  = 3.513
  C2 = 10.0 nF
  C4 = 0.20 nF
  R  = 7.51 kΩ

Stage 2:
  Q  = 1.204
  C2 = 10.0 nF
  C4 = 1.73 nF
  R  = 2.57 kΩ

Stage 3:
  Q  = 0.763
  C2 = 10.0 nF
  C4 = 4.29 nF
  R  = 1.63 kΩ
```

```
Stage 4:
  Q  = 0.594
  C2 = 10.0 nF
  C4 = 7.08 nF
  R  = 1.27 kΩ

Stage 5:
  Q  = 0.521
  C2 = 10.0 nF
  C4 = 9.21 nF
  R  = 1.11 kΩ
Stage 6:
  C = 0.13 nF
  R1  = 10.00 kΩ
  R2 = 80.0 kΩ
  K = 8
```





2. Filter Order Selection

When calculating with N=10, the stopband attenuation comes out to approximately 25.22 dB, which only barely satisfies the spec. To ensure a comfortable margin and account for real-world nonidealities, we chose N=11, yielding 28.2 dB attenuation at 20 kHz and ensuring full compliance with the design requirements.

3. Pole Distribution

The 11 Butterworth poles lie on a semicircle in the left-half plane with a radius of 93,536.64 rad/s. Five complex-conjugate pole pairs define the Q values for five second-order Sallen-Key stages, and one real pole is used for the first-order stage. This ensures a flat passband and smooth roll-off.

4. Sallen-Key Implementation Strategy

The 11th-order Butterworth filter was implemented using:

- **Five second-order stages** with Sallen-Key topology (K = 1)
- **One first-order stage** with gain **K = 8**

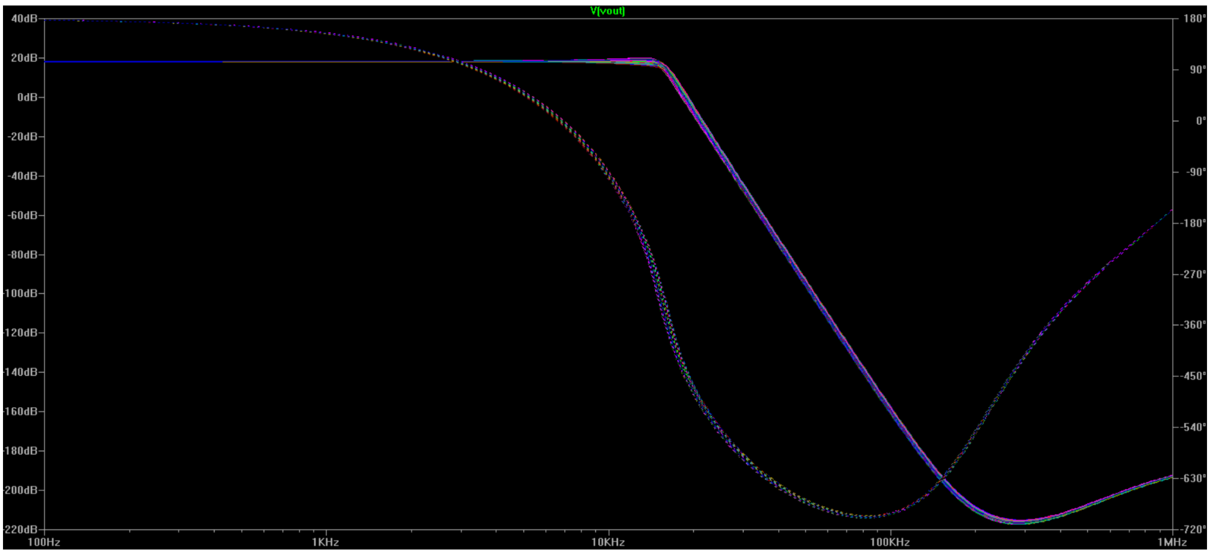
Using the smart design method, each second-order stage fixed **C2 = 10 nF**. The **Q factors** were calculated from the Butterworth poles, and for each stage, component values were derived using:

$$C_4 = \frac{C_2}{4Q^2}, \quad R = \frac{1}{\omega_0\sqrt{C_2C_4}}$$

This ensures consistent ω_0 across all stages while meeting the required Q values.

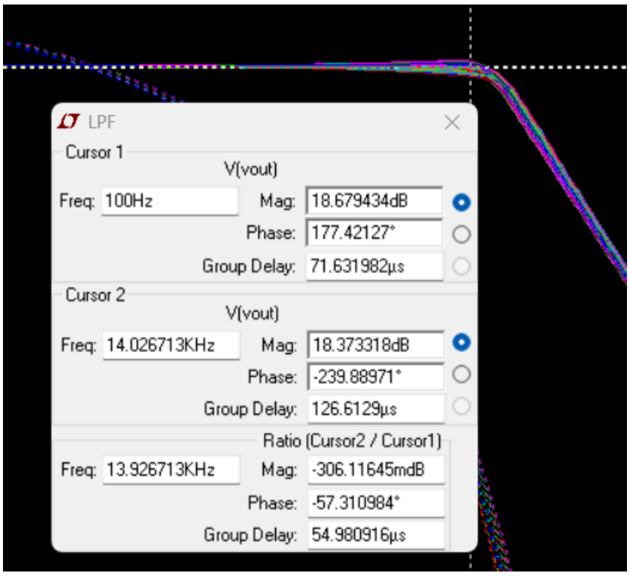
5. Monte Carlo Tolerance Analysis

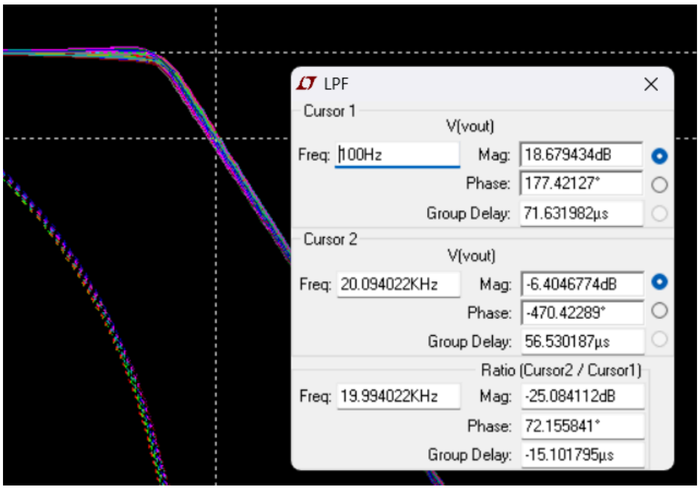
This LTspice code runs a **Monte Carlo AC analysis** with 500 iterations to test how component tolerances affect the frequency response. The `uni(nom, tol)` function randomly perturbs component values, and `.ac dec 100 100` 1Meg sweeps the gain from 100 Hz to 1 MHz for each run.



6. Frequency Response Verification

The designed filter achieves a total gain of **18.67 dB (~8.5×)**. At **20 kHz**, the attenuation is **–25.08 dB**, satisfying the **≥25 dB** stopband spec. The passband ripple at **14 kHz** is **–0.31 dB**, comfortably within the **1 dB limit**. Therefore, the filter fully meets both ripple and attenuation requirements.





8. Comparison Table

Specification	Required	Achieved	Verified
DC Gain	18.06 dB (≈8×)	18.67 dB (≈8.5×)	✅ Yes
PB Ripple	≤ 1 dB	0.31 dB	✅ Yes
SB Attenuation	≥ 25 dB	25.08 dB	✅ Yes

☐ 7. Cutoff Frequency Adjustment and Attenuation Verification

Starting from the specifications of **14 kHz passband edge (fp)** and **20 kHz stopband edge (fs)**, the Butterworth design procedure yielded a natural cutoff frequency of
ω_0 = 93,536.64 rad/s, which corresponds to **f_0 ≈ 14.85 kHz**.

This frequency (14.85 kHz) is slightly higher than the original f□, because the Butterworth filter's definition places its –3 dB point at the natural frequency. At this point, the gain is about **1 dB below the DC gain**, which aligns with the **Amax = 1 dB** specification.

Due to the shift in f□ to 14.85 kHz, the stopband frequency scales accordingly:
fs effective ≈ 21.21 kHz (i.e., 20 kHz scaled by the same ratio).

At this new stopband edge, the calculated attenuation is approximately **28.2 dB**, which safely exceeds the required **25 dB** attenuation.